

미적분13주차 과제

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1. $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1}}{(n+1)!}}{\frac{(-1)^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$ 6022492 완성함
 비한정법 4점

2. $\sum_{n=1}^{\infty} \frac{n-1}{2n!} + \sum_{n=2}^{\infty} \frac{1}{n \ln n}$ 비한정법
 $\lim_{n \rightarrow \infty} \frac{n-1}{2n!} = \frac{1}{2} \neq 0$ 비한정 $\int_0^{\infty} \frac{1}{x \ln x} dx = \infty$ 비한정

3. $\lim_{n \rightarrow \infty} \left| \frac{\log^{(n+1)} \ln \left(1 + \frac{1}{n!} \right) x^{n+1}}{\log^{(n)} \ln \left(1 + \frac{1}{n!} \right) x^n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{n!}} \cdot \frac{-2}{(n!)^2}}{\frac{1}{1 + \frac{1}{n!}} \cdot \frac{2}{n!}} = \lim_{n \rightarrow \infty} \frac{\ln(\ln 2)}{\frac{2}{n!}} = \lim_{n \rightarrow \infty} \frac{\ln(\ln 2)}{2} = \frac{\ln(\ln 2)}{2} < 1$
 4점/비한정 = 1 비한정법

$x=1$ $\sum_{n=1}^{\infty} (-1)^n \ln \left(1 + \frac{1}{n!} \right)$ $\lim_{n \rightarrow \infty} (-1)^n \ln \left(1 + \frac{1}{n!} \right) = 0$ 4점
 극한비교판정법

$x=-1$ $\sum_{n=1}^{\infty} \ln \left(1 + \frac{1}{n!} \right)$ 극한비교판정법 4점

$-1 \leq x \leq 1$